

Data Analytics Internship

Final Executive Summary Report | ApexPlanet Software Pvt. Ltd.

Executive Summary

Over the course of this 30-day internship, I completed an end-to-end data analytics project on the Superstore Sales dataset. The work spanned data cleaning and exploratory analysis, SQL-based data extraction, interactive dashboarding in Power BI, statistical hypothesis testing, time series analysis, customer segmentation via K-Means clustering, and a predictive sales model using Linear Regression. This report summarizes the key findings and business recommendations derived from that analysis.

Project Scope

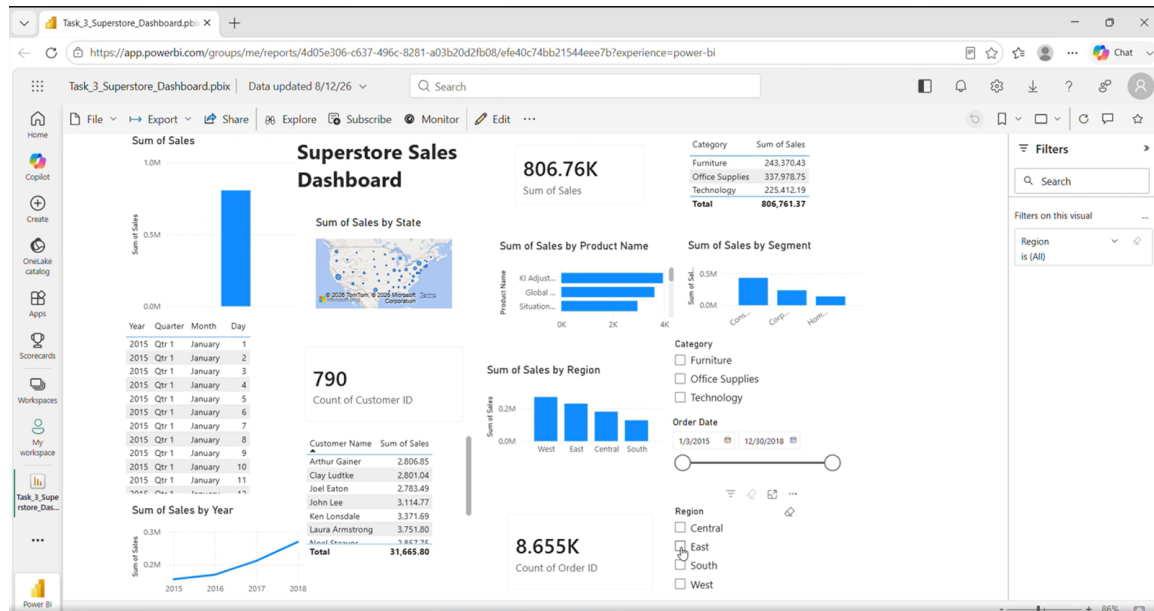
Task	Focus Area	Key Deliverable
Task 1	EDA & Data Cleaning	Cleaned dataset, statistical summaries, visualizations
Task 2	SQL for Data Extraction	SQL queries, Python-SQLite integration
Task 3	Visualization & Dashboarding	Power BI executive dashboard
Task 4	Advanced Analytics	Hypothesis tests, clustering, prediction model
Task 5	Final Report & Automation	This report + automated data pipeline

Top 5 Insights

- Sales Growth & Seasonality:** Overall sales show a consistent upward trend, with a strong recurring spike every November-December, indicating a holiday shopping pattern.
- Regional Sales Parity:** A t-test comparing South and West region sales found no statistically significant difference ($p = 0.09$), suggesting regional performance is broadly comparable.
- Category-Region Independence:** A chi-square test showed no significant relationship between product category and region ($p = 0.75$) — category mix does not vary meaningfully by location.
- Distinct Customer Segments:** K-Means clustering identified 4 customer groups, ranging from high-value "VIP" customers (avg. \$2,122 spend) to occasional low-frequency buyers (avg. \$393 spend).
- Product Drivers of Sales:** Copiers, Machines, and Tables are the strongest positive drivers of sales, while Fasteners, Furnishings, and Office Supplies are associated with lower sales values.

Dashboard Preview

Executive dashboard built in Power BI, showing total sales, sales by state/region/product/segment, customer counts, and order counts.



Business Recommendations

- 1. Launch a VIP loyalty program.** Target Cluster 0 customers (highest spend and order frequency) with early access to promotions and loyalty rewards to strengthen retention among the most valuable segment.
- 2. Prepare inventory ahead of the holiday season.** Given the consistent November-December sales spike, increase stock and marketing focus on Copiers, Machines, and Tables in the lead-up to Q4.
- 3. Run win-back campaigns for low-engagement customers.** Cluster 3 customers show the lowest spend and order frequency; targeted re-engagement offers could recover this segment's contribution to revenue.